

AIRPORT	
<u>AirportCode</u>	PK
Name	
City	
State	

AIRPLANE_TYPE	
<u>TypeName</u>	PK
Company	
MaxSeats	

AIRPORT_AIRPLANE_TYPE	
<u>AirportCode</u>	PK, FK -> AIRPORT
<u>TypeName</u>	PK, FK -> AIRPLANE_TYPE

AIRPLANE	
<u>AirplaneID</u>	PK
TotalSeats	
TypeName	FK -> AIRPLANE_TYPE

FLIGHT	
<u>FlightNo</u>	PK
Airline	
Restrictions	

FLIGHT_WEEKDAY	
<u>FlightNo</u>	PK, FK -> FLIGHT
<u>Weekday</u>	PK

FLIGHT_LEG	
<u>LegNo</u>	PK
FlightNo	FK -> FLIGHT
DepartureAirportCode	FK -> AIRPORT
ArrivalAirportCode	FK -> AIRPORT
ScheduledDepartureTime	
ScheduledArrivalTime	

LEG_INSTANCE	
<u>LegNo</u>	PK, FK -> FLIGHT_LEG
<u>InstanceDate</u>	PK, partial key
AirplaneID	FK -> AIRPLANE
ActualDepartureTime	
ActualArrivalTime	
AvailableSeats	

FARE	
<u>FlightNo</u>	PK, FK -> FLIGHT
<u>FareCode</u>	PK, partial key
Amount	

RESERVATION	
<u>LegNo</u>	PK, composite FK -> LEG_INSTANCE
<u>InstanceDate</u>	PK, composite FK -> LEG_INSTANCE
<u>SeatNo</u>	PK, partial key
CustomerName	
CustomerPhone	

Mapping decisions

M:N ALLOWS_LANDING becomes AIRPORT_AIRPLANE_TYPE; multivalued Weekdays becomes FLIGHT_WEEKDAY.
LEG_INSTANCE, FARE and RESERVATION use owner keys plus their partial keys.

College Management System - Relational Schema

Sprint 3 - ERD to Relational Mapping

System 2 of 3

DEPARTMENT

DepartmentID PK
DName

FACULTY

FacultyID PK
Name
MobileNo
Salary
DepartmentID FK -> DEPARTMENT

STUDENT

StudentID PK
FName
LName
PhoneNo
DOB
DepartmentID FK -> DEPARTMENT
HostelID FK -> HOSTEL, nullable

FACULTY_STUDENT

FacultyID PK, FK -> FACULTY
StudentID PK, FK -> STUDENT

HOSTEL

HostelID PK
HostelName
Address
City
State
PinCode
NoOfSeats

COURSE

CourseID PK
CourseName
Duration
DepartmentID FK -> DEPARTMENT

SUBJECT

SubjectID PK
SubjectName

STUDENT_COURSE

StudentID PK, FK -> STUDENT
CourseID PK, FK -> COURSE

EXAM

ExamCode PK
ExamDate
ExamTime
Room
DepartmentID FK -> DEPARTMENT

FACULTY_SUBJECT

FacultyID PK, FK -> FACULTY
SubjectID PK, FK -> SUBJECT

STUDENT_SUBJECT

StudentID PK, FK -> STUDENT
SubjectID PK, FK -> SUBJECT

STUDENT_EXAM

StudentID PK, FK -> STUDENT
ExamCode PK, FK -> EXAM

Mapping decisions

Student Name is flattened to FName and LName; Age is derived from DOB and is not stored.

All M:N teaching, enrolment, subject and exam relationships use separate junction tables.

EMPLOYEE	
<u>SSN</u>	PK
FName	
Minit	
LName	
Address	
Sex	
BDate	
Salary	
DepartmentName	FK -> DEPARTMENT
StartDate	
SupervisorSSN	FK -> EMPLOYEE, nullable

DEPARTMENT	
<u>DepartmentName</u>	PK
DepartmentNumber	AK, UNIQUE
ManagerSSN	FK -> EMPLOYEE, UNIQUE

DEPARTMENT_LOCATION	
<u>DepartmentName</u>	PK, FK -> DEPARTMENT
<u>Location</u>	PK

DEPENDENT	
<u>EmployeeSSN</u>	PK, FK -> EMPLOYEE
<u>DependentName</u>	PK, partial key
Sex	
BirthDate	
Relationship	

PROJECT	
<u>ProjectName</u>	PK
<u>ProjectNumber</u>	PK
Location	
DepartmentName	FK -> DEPARTMENT

WORKS_ON	
<u>EmployeeSSN</u>	PK, FK -> EMPLOYEE
<u>ProjectName</u>	PK, composite FK -> PROJECT
<u>ProjectNumber</u>	PK, composite FK -> PROJECT
Hours	relationship attribute

Mapping decisions

SupervisorSSN is a self-referencing FK; ManagerSSN maps the 1:1 MANAGES relationship on the total side.
Locations gets its own table; DEPENDENT is weak; WORKS_ON keeps the relationship attribute Hours.